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Class : 12

Register Number

COMMON QUARTERLY EXAMINATION - 2025-26

Time Allowed : 3.00 Hours] PHYSICS [Max. Marks : 70

PART - I

I. **Choose the correct answer.** 15x1=15

1. A temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C its resistance is 1Ω the resistance of the wire will be 2Ω at
   a) 800 °C b) 700 °C c) 850 °C d) 820 °C
2. If the charge on a capacitor is doubled, and voltage applied on it is increased from V to 4 V, then
   a) C increases by Two times b) C decreases by Two times
   c) C increases by four times d) C remains the Same
3. If an oscillating LC circuit, the maximum charge on the capacitor is Q. The charge on the capacitor when energy is stored equally between electric and Magnetic Fields is
   a) $Q/2$ b) $Q/\sqrt{3}$ c) $Q/\sqrt{2}$ d) Q
4. If $E = E_0 \sin [10^5 x - \omega t]$ be the electric field of a plane electromagnetic wave, the value of ω is
   a) 0.3×10^{-14} rad S^{-1} b) 3×10^{-14} rad S^{-1} c) 0.3×10^{14} rad S^{-1} d) 3×10^{14} rad S^{-1}
5. The radius of curvature of curved surface at a thin plane convex lens is 10 cm and the refractive index is 1.5. If the plane surface is silvered, then the focal length will be
   a) 5 cm b) 10 cm c) 15 cm d) 20 cm
6. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole. Moment of the coil is nearly.
   a) 1.0 Am^2 b) 1.2 Am^2 c) 0.5 Am^2 d) 0.2 Am^2
7. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the Bigger sphere is,
   a) 3×10^{-2}C b) 4×10^{-2}C c) 1×10^{-2}C d) 2×10^{-2}C
8. Negative thomson effect is observed in
   a) Silver b) Platinum c) Zinc d) Cadmium
9. The magnetic flux through a surface area $\vec{A}$ in a uniform magnetic field B is maximum, When
   a) θ=0° b) θ=45° c) θ=90° d) θ=270°
10. If the focal length of a lens, is 50 cm, the power of lens is
    a) 1 D b) 0.5 D c) 2 D d) 1.5 D
11. A circular coil with a cross - sectional area of 4 cm^2 has 10 turns. It is placed at the centre of a long solenoid that has 15 turns/ cm and a cross - sectional area of 10 cm^2. The axis of the coil coincides with the axis of the solenoid. What is their mutual inductance?
    a) 7.54 μH b) 8.54 μH c) 9.54 μH d) 10.54 μH
12. Two identical conducting balls having positive charges q_1 and q_2 are separated by a centre to centre distance *r*. If they are made to touch each other and then separated to the same distance, the force between then will be
    a) Less than before b) Same as before c) More than before d) Zero
13. In a large building there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 Heater of 1KW are connected. The voltage of electric mains is 220 V. The maximum capacity of the main fuse of the building Will be
    a) 14 A b) 8 A c) 10 A d) 12 A
14. Star twinkle due to
    a) Reflection b) Total internal reflection
    c) Refraction d) Polarisation
15. Copper less in transformer can be minimized by
    a) Winding coils one over the other b) Using steel of high Silicon content
    c) Using thin laminations of transformer core d) Using wires of larger diameter

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PART - II

II. Answer Any SIX of The Following. (Answer Question No. 24 Compulsory) 6x2=12

16 State Fleming's left hand Rule

17 What is the reason for reddish appearance of sky during sunset and sunrise?

18 When two objects are rubbed with each other, approximately a charge of 50 n C can be produced in each object. Calculate the number of electrons that must be transferred to produce this charge.

19 Define Electrical Resistivity

20 What are the properties of Equipotential surface?

21 Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at a distance 1 m from it

22 What are Phasors?

23 Give two uses of UV radiation.

24 If the relative permeability and relative permittivity of the medium are 1.0 and 2.25, respectively. Find the speed of the electromagnetic wave in this medium.

PART - III

III. Answer Any SIX of The Following. (Answer Question No.33 Compulsory) 6x3=18

25. Give the applications of Seebeck effect?

26. An inductor of inductance L carries an electric current i. how much energy is stored while establishing the current in it?

27. An object is placed at a certain distance from a convex lens of focal length 20 cm. Find the object distance if the image obtained is. magnified 4 times.

28. Derive the expression for resultant capacitance when capacitors are connected in series

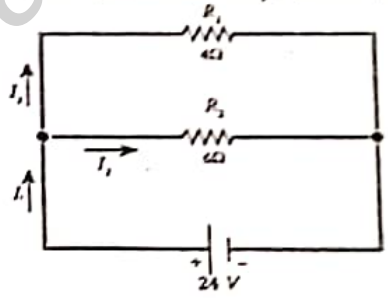
29. Write down Maxwell's equations in integral form.

30. Find the impedance of a series of RLC circuit if the inductive reactance, Capacitive reactance and resistance , are 184 Ω , 144 Ω and 30 Ω respectively. Also calculate the phase angle between voltage and current.

31. Define current sensitively of a galvanometer. How will you increase the current sensitivity of a Galvanometer.

32. Derive the relation between *f* and R for a spherical mirror.

33. Calculate the equivalent resistance in the following circuit and also find the values of current, I, I_1 and I_2 in the given circuit.



PART - IV

IV. Answer ALL Questions. Draw the diagrams wherever necessary. 5x5=25

34. a) Find out the phase relationship between Voltage and Current in a pure inductive circuit.

(OR)

b) Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one Cycle.

35. a) Derive the expression for the force between two parallel, current carrying conductors.

(OR)

b) Explain the types of emission spectrum.

36. a) Calculate the electric field due to a dipole on its equatorial plane.

(OR)

b) Obtian the expression for electric field due to an Uniformly charged spherical shell.

37. a) Describe the microscopic model of current and obtain microscopic form of Ohm's law.

(OR)

b) Obtain the condition for bridge balance in Wheatstone's Bridge.

38. a) Derive the mirror Equation. (OR)

b) Derive the equation for refraction at single spherical surface.

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